



Published in final edited form as:

Med Sci Sports Exerc. 2022 December 01; 54(12): 2208–2215. doi:10.1249/MSS.0000000000003014.

Mechanical Factors Contributing to Altered Knee Extension Moment during Gait after ACL Reconstruction: A Longitudinal Analysis

Chandramouli Krishnan^{1,2,3,4,6}, Alexa K. Johnson², Riann M. Palmieri-Smith^{2,5}

¹Physical Medicine and Rehabilitation, Michigan Medicine, Ann Arbor, MI;

²School of Kinesiology, University of Michigan, Ann Arbor, MI;

³Robotics Institute, University of Michigan, Ann Arbor, MI;

⁴Department of Biomedical Engineering, University of Michigan, Ann Arbor, MI;

⁵Department of Orthopedic Surgery, University of Michigan, Ann Arbor, MI

⁶Physical Therapy Department, College of Health Sciences, University of Michigan-Flint, Flint, MI

Abstract

Purpose: To comprehensively examine the extent to which knee flexion angle at initial contact, peak knee flexion angle, and vertical ground reaction force (vGRF) contribute to knee extension moments during gait in individuals with anterior cruciate ligament (ACL) reconstruction.

Methods: Overground gait biomechanics were evaluated in twenty-six participants with ACL reconstruction at three timepoints (about 2 months, 4 months, and 6 months after the surgery). Knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment were calculated for each limb during the early stance phase of gait for all three timepoints. A change score from baseline (timepoint 2–timepoint 1 and timepoint 3–timepoint 1) along with limb symmetry values (ACL–non-ACL limb values) were also calculated for these variables. Multiple linear regressions utilizing classical and Bayesian interference methods were used to determine the contribution of knee flexion angle and vGRF to knee extension moment during gait.

Results: Peak knee flexion angle and peak vGRF positively contributed to knee extension moment during gait in both the reconstructed ($R^2 = 0.767$, $p < 0.001$) and non-reconstructed limbs ($R^2 = 0.815$, $p < 0.001$). Similar results were observed for the symmetry values ($R^2 = 0.673$, $p < 0.001$) and change scores ($R^2 = 0.731$ – 0.883 ; all $p < 0.001$), except that the changes in knee flexion angle at initial contact also contributed to the model using the change scores in the non-reconstructed limb (timepoint 2–timepoint 1: $R^2 = 0.844$, $p < 0.001$; timepoint 3–timepoint 1:

Address for Correspondence: Chandramouli Krishnan, PT, PhD, University of Michigan, 325 E Eisenhower Parkway, Suite 3013, Ann Arbor, MI – 48108; Phone: (319)-321-0117; Fax: (734)-615-1770; mouli@umich.edu.

Conflict of Interest

None

SUPPLEMENTAL DIGITAL CONTENT

SDC 1: Appendix 1 GRF Knee Moment Bayesian and Linear Regression.pdf – Appendix

$R^2 = 0.883$, $p < 0.001$). Bayesian regression evaluating the likelihood of these prediction models showed that there was decisive evidence favoring the alternative model over the null model (all Bayes Factors $[BF_{10}] > 1000$). Standardized beta coefficients indicated that changes in knee flexion angle had a greater impact ($> 2x$) on knee extension moments than vGRF at both timepoints in both limbs ($\beta_{vGRF} = 0.204-0.309$; $\beta_{kneeflexion} = 0.703-0.831$).

Conclusions: The findings indicate that both knee flexion angle and peak vGRF positively contribute to altered knee extension moments during gait, but the contribution of knee flexion angle is much greater than vGRF. Therefore, treatment strategies targeting these variables may improve knee loading after ACL reconstruction.

Keywords

KNEE MOMENT; KNEE KINETICS; JOINT LOADING; KNEE ANGLE; LIGAMENT INJURY; SPORTS

INTRODUCTION

Restoration of normal gait patterns is a major clinical goal when rehabilitating individuals with anterior cruciate ligament (ACL) reconstruction. However, despite significant rehabilitation efforts to normalize gait patterns, several studies show that altered knee biomechanics is commonplace after ACL reconstruction (1–4). Specifically, knee joint underloading during walking (as measured by knee moments or contact forces) has been identified as a significant issue in these individuals due to the growing evidence that points to its association with the development of post-traumatic osteoarthritis after ACL reconstruction (5–8). These deficits in knee joint loading are seen as early as 2–3 months following surgery and persist for years after the surgery (9–12). Hence, early interventions that target altered knee loading could potentially improve the long-term knee health after ACL reconstruction. However, a critical step before developing new interventions to combat post-ACL reconstruction knee loading deficits is to identify the mechanical factors that contribute to these deficits.

Several mechanical and neuromuscular factors can contribute to altered knee loading during gait. Neuromuscular factors such as quadriceps weakness and dysfunction (e.g., arthrogenic muscle inhibition) may alter knee joint loading *indirectly* by altering the ability of the knee to properly attenuate forces during the loading phase of the gait (13). In contrast, mechanical factors such as knee flexion angle and vertical ground reaction forces (vGRFs) could directly influence knee joint loading, as they serve as primary inputs to the calculation of knee joint moments and contact forces (9). While previous research on the role of neuromuscular factors on knee joint loading has been conflicting (13–15), there has been a consistent link between mechanical factors and knee joint loading. For example, previous research indicates that reduced vertical ground reaction force (vGRF) in the ACL-reconstructed limb is strongly associated with reduced knee extensor moments in the ACL-reconstructed limb and that modulation in vGRF alters knee moments during gait (9, 10). Similarly, there is a moderate positive association between knee flexion angle and knee moments during gait after ACL reconstruction (9). These findings have instigated researchers to use biofeedback to improve knee moments and restore normal gait patterns after ACL reconstruction (16).

While previous research has established the cross-sectional association between vGRF, knee flexion angle, and the knee moment during gait, the longitudinal nature of this relationship is currently unknown. In other words, is the change in vGRF and knee flexion angle with time associated with the change in knee moment with time? Moreover, the extent to which changes in knee angle and vGRF asymmetry between limbs with time contribute to changes in knee moment asymmetry is not clear. Further, prior studies have used classical hypothesis testing based on p-values, which is often affected by small sample size and does not answer the primary research question of interest – what is the probability of the null hypothesis being true given the observed data? These are crucial gaps that need to be filled before implementing any targeted intervention to address altered knee loading after ACL reconstruction. Therefore, the purpose of this study was to comprehensively (both cross-sectionally and longitudinally) examine the extent to which knee flexion angle at initial contact, peak knee flexion angle, and vGRF contribute to knee extension moments during gait in individuals with ACL reconstruction. Participants were evaluated at three timepoints: about 1) 2 months after ACL reconstruction, 2) 4 months after ACL reconstruction, and 3) 6 months after ACL reconstruction. Cross-sectional analyses were performed using the normalized raw data and the symmetry values (ACL limb–non-ACL limb) from all timepoints. Longitudinal analyses were performed at the 4-month and 6-month timepoints using the change scores from baseline (i.e., timepoint 2–timepoint 1 and timepoint 3–timepoint 1). We hypothesized that knee flexion angle at initial contact, peak knee flexion angle, and vGRF would be positively associated with peak knee extension moment and significantly contribute to altered knee loading during gait both cross-sectionally and longitudinally. These hypotheses were evaluated both using the traditional p-values obtained from classical hypothesis testing and Bayes factors obtained from Bayesian statistics.

METHODS

Participants

Twenty-six individuals (17 females, 9 males; Age: 20.7 ± 5.6 years; Height: 1.72 ± 0.10 meters (m), Body mass: 70.0 ± 12.0 kilograms (kg), Body mass index: 23.6 ± 3.6 kg/m²) with ACL reconstruction (21 bone-patellar tendon-bone graft, 4 hamstring tendon graft, and 1 quadriceps tendon graft) volunteered to participate in this study. Participants were part of a clinical trial ([NCT03282565](#)) that examined the effects of functional resistance training on knee strength and biomechanics after ACL reconstruction and were tested at three timepoints: about 1) 2 months after ACL reconstruction (2.2 ± 0.7 months), 2) 4 months after ACL reconstruction (4.6 ± 0.8 months), and 3) 6 months after ACL reconstruction (6.7 ± 0.8 months). Participants in both the intervention and control arms of the study were included to ensure that there was enough spread in the data for regression analyses. Participants were included if they 1) were aged 14–40 years old, 2) had suffered an acute, primary, unilateral complete ACL rupture, and 3) had undergone ACL reconstruction with an autograft. Participants were excluded from the study if they 1) had suffered a previous ACL injury or previous surgery to either knee, 2) had sustained a recent significant knee injury other than the ACL tear, 3) had sustained a bony fracture or knee dislocation accompanying their ACL injury, 4) were pregnant or planning to become pregnant, 5) had a medical condition (e.g.,

a history of uncontrolled diabetes or hypertension) that could negatively affect the study results, and/or 6) were unable to provide written informed consent. All participants read and signed an informed consent/assent document approved by the University of Michigan Medical Institutional Review Board before their participation. Parental consent was obtained if the participant was a minor child.

Gait Biomechanics Testing

Overground knee biomechanical gait data were collected using a 15-camera motion capture system (MX-13, Vicon Motion System Ltd., Oxford, GBR) and two independent force plates (AMTI OR6–7, Advanced Mechanical Technology Inc., Watertown, MA, USA) embedded on level ground. Motion capture data (240Hz) were synchronized with GRF data (1200Hz) using Vicon hardware/software. A total of 34 reflective markers (14 mm diameter) were placed bilaterally over the participant's anatomical locations (17). A static trial of each participant aligned within the laboratory coordinate system was recorded, following which walking trials were performed. Five trials were collected at the participant's self-selected gait speed during all three testing sessions and were used in further analysis. Participants were instructed to walk at their comfortable, habitual walking pace. Marker data and ground reaction forces were filtered with a zero phase-lag low-pass Butterworth filter (4th order, 12 Hz cut-off). Visual 3D software (v3.90, C-Motion, Rockville, MD, USA) was used to calculate bilateral knee sagittal plane flexion-extension motions and internal moments. Lower limb joint rotations were calculated using the Cardan rotation sequence and were expressed relative to each participant's neutral static position recorded during the static trial. Joint moments were normalized to the participant's height and mass (obtained at each timepoint) and vGRF was normalized to the participant's body weight. All biomechanical data were time normalized to the stance phase of the gait cycle, with the initial contact and toe-off being defined using a vGRF threshold of 10 N (18). The ensemble averages of the time-normalized knee angle, vGRF, and knee moment data were used in further analyses. Knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment were calculated for each limb during the early stance phase of gait, identified as the first 50% of the stance phase, for all three timepoints (Figure 1) (15). We chose the first 50% of the stance phase for two reasons: (1) Most of the prior research in ACL-reconstructed population has almost exclusively focused on the peak knee moment during the loading response/early stance, and (2) this is also the region where individuals with ACL reconstruction typically exhibit differences in gait mechanics when compared with uninjured controls. A change score (timepoint 2–timepoint 1 and timepoint 3–timepoint 1) along with limb symmetry values (*i.e.*, ACL limb values–non-ACL limb values) were also calculated for these variables.

Statistical Analyses

All statistical analyses were performed using the JASP version 0.14.0.0 statistical software (<https://jasp-stats.org/>), and the JASP outputs are provided in Appendix 1 and in the Open Science Framework (OSF - <https://osf.io/zxkes/>). Descriptive statistics were computed for each variable, and rain cloud plots were generated for robust data visualization (19). Stepwise multiple linear regression analysis was used to determine the contribution of knee flexion angle at initial contact, peak knee flexion angle during early stance, and

peak vGRF during early stance to knee joint loading (i.e., peak knee moment) during the early stance phase of gait. Separate regression analyses were performed using the data from the reconstructed limb, non-reconstructed limb, symmetry values, and the change scores to comprehensively understand the contributions of knee flexion angle and vGRF to peak knee extension moment during gait. We adopted an inclusive statistical approach where both classical and Bayesian regression analyses were performed to measure statistical significance (p-value) and the Bayes Factor (BF₁₀), respectively (20–25). More details regarding the Bayesian analysis and how to interpret the results of this analysis are provided in the JASP manual: (http://static.jasp-stats.org/Manuals/Bayesian_Guide_v0_12_2_1.pdf). Briefly, Bayesian regression analyses in JASP were conducted using the default prior to compute BF₁₀ that describes the likelihood of the alternative model ($\beta \neq 0$) to the likelihood of the null model ($\beta = 0$). Thus, the BF₁₀ indicates the relative strength of evidence in the data for two competing hypotheses. We used the descriptive classifications adopted by JASP (based Jeffreys's scheme (26)) to identify Bayes Factor values as none, anecdotal, moderate, strong, very strong, or decisive relative evidence for our alternative hypothesis (Table 1). The Volk-Selke Maximum p-Ratio (VS-MPR) was also computed to estimate the maximum possible odds in favor of the alternative hypothesis (H₁) over the null hypothesis (H₀), based on a two-sided p-value (27, 28). This was computed using the following equation:

$$\text{VS-MPR} = \frac{1}{[-e \times p \times \ln(p)]} \text{ for } p \leq 0.37$$

A significance level of $\alpha = 0.05$ was used for classical regression analyses.

RESULTS

Raw Scores

Raincloud plots showing the distributions of knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment for the ACL-reconstructed and non-ACL-reconstructed limbs at the three timepoints are shown in Figure 2. Regression analyses indicated that peak knee flexion angle and peak vGRF together significantly predicted peak knee extension moment during gait in both the reconstructed ($p < 0.001$, $R^2 = 0.767$, VS-MPR = $1.702e + 21$; Figure 3A) and the non-reconstructed limbs ($p < 0.001$, $R^2 = 0.815$, VS-MPR = $7.616e + 24$; Figure 3B). Bayesian regression evaluating the likelihood of these prediction models showed that there was decisive evidence favoring the alternative regression model that included peak knee flexion angle and peak vGRF as opposed to the null model that only included the intercept (ACL BF₁₀ = $8.463e + 20$ and non-ACL BF₁₀ = $3.607e + 24$).

Limb Symmetry Values

Raincloud plots showing the distributions of limb symmetry values for knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment at the three timepoints are shown in Figure 4. Regression analyses indicated that peak knee flexion angle symmetry and peak vGRF symmetry together significantly predicted peak knee extension moment symmetry during gait ($p < 0.001$, $R^2 = 0.673$, VS-MPR =

8.197e +15; Figure 3C). Bayesian regression evaluating the likelihood of these prediction models showed that there was decisive evidence favoring the alternative regression model that included peak knee flexion angle symmetry and peak vGRF symmetry as opposed to the null model that only included the intercept ($BF_{10} = 4.390e +15$).

Change Scores

Raincloud plots showing the distributions of changes (from timepoint 1) in knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment for the ACL-reconstructed and non-ACL-reconstructed limbs at timepoints 2 and 3 are shown in Figure 5. When evaluating the reconstructed limb, regression analyses indicated that the changes in peak knee flexion angle and peak vGRF together significantly predicted the changes in peak knee extension moment during gait at both timepoint 2 ($p < 0.001$, $R^2 = 0.775$, VS-MPR = 606628.184; Figure 6A) and timepoint 3 ($p < 0.001$, $R^2 = 0.731$, VS-MPR = 47973.267; Figure 6C). Bayesian regression evaluating the likelihood of these prediction models showed that there was decisive evidence favoring the alternative regression model that included changes in peak knee flexion angle and peak vGRF as opposed to the null model that only included the intercept (ACL BF_{10} at timepoint 2 = 329729.448 and ACL BF_{10} at timepoint 3 = 27481.216). When evaluating the non-reconstructed limb, regression analyses indicated that the changes in knee flexion angle at initial contact, peak knee flexion angle, and peak vGRF together significantly predicted the changes in peak knee extension moment during gait at both timepoint 2 ($p < 0.001$, $R^2 = 0.844$, VS-MPR = 4.129e +6; Figure 6B) and timepoint 3 ($p < 0.001$, $R^2 = 0.883$, VS-MPR = 3.068e +7; Figure 6D). Bayesian regression evaluating the likelihood of these prediction models showed that there was decisive evidence favoring the regression model that included changes in peak knee flexion angle and peak vGRF as opposed to the null model that only included the intercept (non-ACL BF_{10} at timepoint 2 = 2.526e +6 and non-ACL BF_{10} at timepoint 3 = 1.779e +7).

DISCUSSION

This study was performed to evaluate the extent to which knee flexion angle and vGRF contribute to knee extension moment during walking in individuals with ACL reconstruction. As hypothesized, both peak knee flexion angle and peak vGRF significantly contributed to peak knee extension moment during the early stance phase of gait. Specifically, these variables explained more than 75% of the variance in the peak knee extension moment data in both the ACL-reconstructed and the non-ACL-reconstructed limbs. While the amount of variance explained by the model reduced slightly when using the symmetry values, peak knee flexion angle and peak vGRF symmetry values still explained more than 67% of the variance in peak knee extension moment symmetry. More importantly, the longitudinal changes in peak knee flexion angle and peak vGRF explained about 73% to 84% of the variance in peak knee extension moment changes after ACL reconstruction. The Bayes factors (BF_{10}) and VS-MPR values for each of these predictive models were also extremely high ($>10,000$), indicating that the evidence in favor of the alternative predictive models over the null model was decisive. Collectively, these findings indicate that alterations in knee kinematics (peak knee flexion angle) and kinetics (vGRF) contribute to altered knee

loading (knee extension moments) during gait and emphasize the importance of developing targeted interventions that improve knee kinematics and kinetics to restore normal gait mechanics after ACL reconstruction.

Individuals with ACL reconstruction exhibit abnormal gait patterns (e.g., reduced GRFs and knee flexion angle) that do not recover even years after surgery and rehabilitation (1–4). These altered gait patterns have been shown to contribute to abnormal knee loading (e.g., reduced knee moment) and poor knee function after ACL reconstruction and have been linked to the pathogenesis of posttraumatic osteoarthritis (5, 6, 9, 10, 29, 30). Accordingly, researchers have placed considerable efforts to develop new interventions targeting abnormal knee loading in individuals with ACL reconstruction. However, a clear understanding of the mechanical factors that contribute to altered knee loading after ACL reconstruction is critical to developing appropriate intervention strategies. The results of this study indicate that peak knee flexion angle and peak vGRF during early stance contribute significantly to the peak knee extension moment during gait, which is consistent with previous cross-sectional research showing that reduced GRFs and knee flexion range of motion are strongly associated with reduced knee extensor moments 3–4 months after ACL reconstruction (9). Interestingly, when evaluating the standardized coefficients, it appears that the peak knee flexion angle contributes to a much larger extent ($>2\times$) than the peak vGRF to the observed knee extension moments during gait (see Appendix, Supplemental Digital Content 1). This finding suggests that interventions that target to increase knee flexion angles may have a higher impact on improving knee joint loading than interventions that target to increase vGRF. Knee flexion angles are also easier to target in a clinic due to the increasing availability of low-cost camera systems that can provide real-time feedback of knee joint angles (31–33). However, further research is warranted to determine if targeting greater knee flexion angle during gait can improve knee joint loading and cartilage health following ACL reconstruction.

There are two novel aspects to this study. First, unlike previous studies, we performed a comprehensive analysis of the relationship between vGRF, knee flexion angle, and peak knee moment. Specifically, we evaluated the ability of peak vGRF and peak knee flexion angle to predict peak knee extension moment cross-sectionally using individual limb data and symmetry values. We also examined the relationships between these variables longitudinally to understand how a change in one variable contributes to the other. This information is critical because a strong initial correlation between vGRF, knee flexion angle, and knee moment does not necessarily imply that these variables will also follow a similar trend of improvement with time (or training), which can be determined using a change/gain score (i.e., changes across timepoints) (34). Here, using a change score, we show for the first time that changes in vGRF and peak knee flexion angle predict changes in peak knee extension moment in individuals with ACL reconstruction. These findings suggest that the correlations seen between these variables can be inferred as causative, and hence, interventions improving knee flexion angle and vGRF early after ACL reconstruction could lead to sustained improvements in knee joint loading. Second, we also used a novel Bayesian approach that evaluated the likelihood of these predictive models given the observed experimental data. Unlike traditional hypothesis testing based on p-values, Bayesian analysis provides evidence related to the likelihood of two competing hypotheses

(null and alternative) and allows inferences to be made about whether the observed data is more consistent with the alternative hypothesis or the null hypothesis. The results of this study indicated that the Bayes Factors were very high (>1000), indicating decisive evidence favoring our predictive model over the null model. As indicated before, these are critical findings that strengthen the existing cross-sectional literature and provide a step in the right direction to develop interventions that could potentially address abnormal gait biomechanics after ACL reconstruction.

There are some limitations to this study. One limitation is that we only evaluated the mechanical factors that contribute to peak knee extension moments during gait. However, other factors such as graft type, knee pain, and gait speed can contribute to altered knee moments after ACL reconstruction. We do note that knee angle and vGRF contributed to more than 70% of the variation in peak knee extension moments; thus, the contributions of other variables that were not included in the analyses must be only minimal. Another limitation is that we only evaluated the peak knee extension moments during the first 50% of the stance phase. This was an intentional choice because previous studies have primarily focused on peak knee moments during the loading response/early stance phase of the gait, where individuals with ACL reconstruction exhibit differences in gait mechanics when compared with uninjured controls. However, it is likely that knee moments during the latter part of the stance phase may also be crucial to the development of post-traumatic knee osteoarthritis and need to be evaluated in future studies. Finally, while we evaluated the causal association of knee flexion angle and vGRF with knee extension moment during gait, we do not know whether improvements in knee flexion angle and vGRF would lead to sustained improvements in knee moments during gait. Future studies are warranted to examine this important research question, as it has implications for improving gait function after ACL reconstruction.

CONCLUSIONS

In conclusion, the results of this study indicated that peak knee flexion angle and vGRF significantly predicted peak knee extension moment in individuals with ACL reconstruction, both cross-sectionally and longitudinally. Specifically, cross-sectional analyses using the normalized raw values and symmetry values from all timepoints indicated that peak knee flexion angle and peak vGRF together contributed positively to knee extension moment during gait. In addition, longitudinal analyses using change scores from baseline indicated that changes in peak knee flexion angle and vGRF with time predicted changes in peak knee extension moment, with knee flexion angle influencing knee extension moment more than vGRF in both limbs at both timepoints. Bayesian regression indicated that there was decisive evidence favoring the likelihood of these prediction models. These results suggest that treatment strategies targeting greater knee flexion angle and vGRF may improve knee loading after ACL reconstruction.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

This research was supported by the National Institute of Health NICHD (R21-HD092614). The content is solely the responsibility of the authors and does not represent the views of the NIH. The authors would also like to thank the past and current members of the NeuRRo Lab and the ORB Lab for their assistance with data collection and processing. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

Conflict of Interest and Funding Source:

This research was supported by the National Institute of Health NICHD (R21-HD092614). The authors declare no conflicts of interest. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

REFERENCES

1. Capin JJ, Khandha A, Zarzycki R, Manal K, Buchanan TS, Snyder-Mackler L. Gait mechanics and second ACL rupture: implications for delaying return-to-sport. *J Orthop Res.* 2017;35(9):1894–901. [PubMed: 27859527]
2. Hart HF, Culvenor AG, Collins NJ, et al. Knee kinematics and joint moments during gait following anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *Br J Sports Med.* 2016;50(10):597–612. [PubMed: 26265562]
3. Kaur M, Ribeiro DC, Theis JC, Webster KE, Sole G. Movement patterns of the knee during gait following ACL reconstruction: a systematic review and meta-analysis. *Sports Med.* 2016;46(12):1869–95. [PubMed: 26936269]
4. Slater LV, Hart JM, Kelly AR, Kuenze CM. Progressive changes in walking kinematics and kinetics after anterior cruciate ligament injury and reconstruction: a review and meta-analysis. *J Athl Train.* 2017;52(9):847–60. [PubMed: 28985125]
5. Amano K, Podoia V, Su F, Souza RB, Li X, Ma CB. Persistent biomechanical alterations after ACL reconstruction are associated with early cartilage matrix changes detected by quantitative MR. *Orthop J Sports Med.* 2016;4(4):2325967116644421.
6. Li H, Tao H, Hua Y, Chen J, Li Y, Chen S. Quantitative magnetic resonance imaging assessment of cartilage status: a comparison between young men with and without anterior cruciate ligament reconstruction. *Arthroscopy.* 2013;29(12):2012–9. [PubMed: 24286800]
7. Shimizu T, Samaan MA, Tanaka MS, et al. Abnormal biomechanics at 6 months are associated with cartilage degeneration at 3 years after anterior cruciate ligament reconstruction. *Arthroscopy.* 2019;35(2):511–20. [PubMed: 30473456]
8. Culvenor AG, Perraton L, Guermazi A, et al. Knee kinematics and kinetics are associated with early patellofemoral osteoarthritis following anterior cruciate ligament reconstruction. *Osteoarthritis Cartilage.* 2016;24(9):1548–53. [PubMed: 27188685]
9. Lin PE, Sigward SM. Contributors to knee loading deficits during gait in individuals following anterior cruciate ligament reconstruction. *Gait Posture.* 2018;66:83–7. [PubMed: 30170138]
10. Sigward SM, Lin P, Pratt K. Knee loading asymmetries during gait and running in early rehabilitation following anterior cruciate ligament reconstruction: a longitudinal study. *Clin Biomech (Bristol, Avon).* 2016;32:249–54.
11. Capin JJ, Zarzycki R, Arundale A, Cummer K, Snyder-Mackler L. Report of the primary outcomes for gait mechanics in men of the ACL-SPORTS trial: secondary prevention with and without perturbation training does not restore gait symmetry in men 1 or 2 years after ACL reconstruction. *Clin Orthop Relat Res.* 2017;475(10):2513–22. [PubMed: 28224442]
12. Garcia SA, Brown SR, Koje M, Krishnan C, Palmieri-Smith RM. Gait asymmetries are exacerbated at faster walking speeds in individuals with acute anterior cruciate ligament reconstruction. *J Orthop Res.* 2022;40(1):219–30. [PubMed: 34101887]

13. Arhos EK, Capin JJ, Buchanan TS, Snyder-Mackler L. Quadriceps strength symmetry does not modify gait mechanics after anterior cruciate ligament reconstruction, rehabilitation, and return-to-sport training. *Am J Sports Med.* 2021;49(2):417–25. [PubMed: 33373534]
14. Di Stasi SL, Logerstedt D, Gardinier ES, Snyder-Mackler L. Gait patterns differ between ACL-reconstructed athletes who pass return-to-sport criteria and those who fail. *Am J Sports Med.* 2013;41(6):1310–8. [PubMed: 23562809]
15. Lewek M, Rudolph K, Axe M, Snyder-Mackler L. The effect of insufficient quadriceps strength on gait after anterior cruciate ligament reconstruction. *Clin Biomech (Bristol, Avon).* 2002;17(1):56–63.
16. Evans-Pickett A, Davis-Wilson HC, Luc-Harkey BA, et al. Biomechanical effects of manipulating peak vertical ground reaction force throughout gait in individuals 6–12 months after anterior cruciate ligament reconstruction. *Clin Biomech (Bristol, Avon).* 2020;76:105014.
17. Brown SR, Washabaugh EP, Dutt-Mazumder A, Wojtys EM, Palmieri-Smith RM, Krishnan C. Functional resistance training to improve knee strength and function after acute anterior cruciate ligament reconstruction: a case study. *Sports Health.* 2021;13(2):136–44. [PubMed: 33337984]
18. Lepley AS, Gribble PA, Thomas AC, Tevald MA, Sohn DH, Pietrosimone BG. longitudinal evaluation of stair walking biomechanics in patients with ACL injury. *Med Sci Sports Exerc.* 2016;48(1):7–15. [PubMed: 26225766]
19. Allen M, Poggiali D, Whitaker K, Marshall TR, Kievit RA. Raincloud plots: a multi-platform tool for robust data visualization. *Wellcome Open Res.* 2019;4:63. [PubMed: 31069261]
20. Wagenmakers EJ, Love J, Marsman M, et al. Bayesian inference for psychology. Part II: example applications with JASP. *Psychon Bull Rev.* 2018;25(1):58–76. [PubMed: 28685272]
21. Wagenmakers EJ, Marsman M, Jamil T, et al. Bayesian inference for psychology. Part I: theoretical advantages and practical ramifications. *Psychon Bull Rev.* 2018;25(1):35–57. [PubMed: 28779455]
22. Gardi AZ, Vogel AK, Dharia AK, Krishnan C. Effect of conventional transcranial direct current stimulation devices and electrode sizes on motor cortical excitability of the quadriceps muscle. *Restor Neurol Neurosci.* 2021;39(5):379–91. [PubMed: 34657855]
23. Joshi V, Rouse EJ, Claflin ES, Krishnan C. How does ankle mechanical stiffness change as a function of muscle activation in standing and during the late stance of walking? *IEEE Trans Biomed Eng.* 2022;69(3):1186–93. [PubMed: 34606446]
24. Krishnan C Effect of paired-pulse stimulus parameters on the two phases of short interval intracortical inhibition in the quadriceps muscle group. *Restor Neurol Neurosci.* 2019;37(4):363–74. [PubMed: 31306142]
25. Ruiz-Ruano Garcia AM, Lopez Puga J. Deciding on null hypotheses using P-values or Bayesian alternatives: a simulation study. *Psicothema.* 2018;30(1):110–5. [PubMed: 29363479]
26. Jeffreys H *Theory of probability* (3rd ed.). Oxford University Press. MR0187257. 1961;432.
27. Sellke T, Bayarri M, Berger JO. Calibration of p values for testing precise null hypotheses. *Am Stat.* 2001;55(1):62–71.
28. Vovk VG. A logic of probability, with application to the foundations of statistics. *J R Stat Soc Series B Stat Methodol.* 1993;55(2):317–41.
29. Erhart-Hledik JC, Chu CR, Asay JL, Andriacchi TP. Gait mechanics 2 years after anterior cruciate ligament reconstruction are associated with longer-term changes in patient-reported outcomes. *J Orthop Res.* 2017;35(3):634–40. [PubMed: 27238273]
30. Khandha A, Manal K, Wellsandt E, Capin J, Snyder-Mackler L, Buchanan TS. Gait mechanics in those with/without medial compartment knee osteoarthritis 5 years after anterior cruciate ligament reconstruction. *J Orthop Res.* 2017;35(3):625–33. [PubMed: 27082166]
31. Krishnan C, Washabaugh EP, Seetharaman Y. A low cost real-time motion tracking approach using webcam technology. *J Biomech.* 2015;48(3):544–8. [PubMed: 2555306]
32. Saner RJ, Washabaugh EP, Krishnan C. Reliable sagittal plane kinematic gait assessments are feasible using low-cost webcam technology. *Gait Posture.* 2017;56:19–23. [PubMed: 28482201]
33. Viswakumar A, Rajagopalan V, Ray T, Gottipati P, Parimi C. Development of a robust, simple, and affordable human gait analysis system using bottom-up pose estimation with a smartphone camera. *Front Physiol.* 2021;12:784865. [PubMed: 35069246]

34. Moreau D, Conway AR. The case for an ecological approach to cognitive training. *Trends Cogn Sci*. 2014;18(7):334–6. [PubMed: 24972505]

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

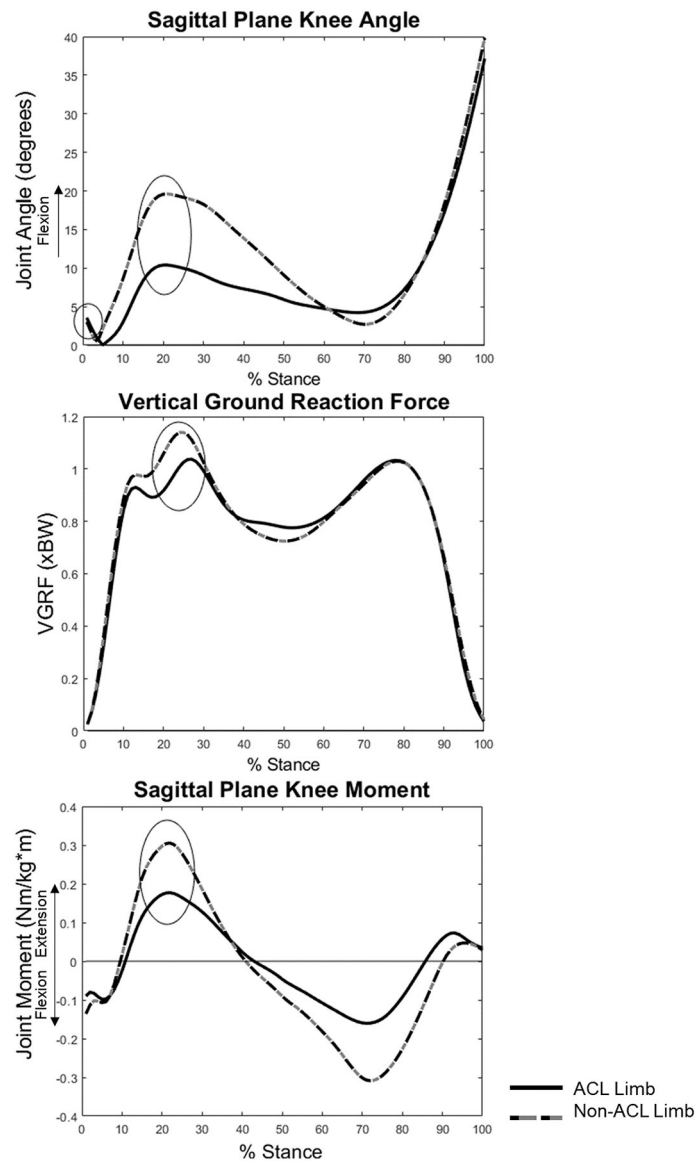


Figure 1:

An exemplary data of sagittal plane knee kinematics, vertical ground reaction force (vGRF), and knee moment from a participant in our study to depict the region at which the analysis was performed. The solid lines represent the anterior cruciate ligament-reconstructed limb (ACL), and the broken lines represent the non-ACL-reconstructed limb (Non-ACL). Ellipses represent the point at which gait data were evaluated: knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment during the early stance phase of gait when the knee is maximally loaded.

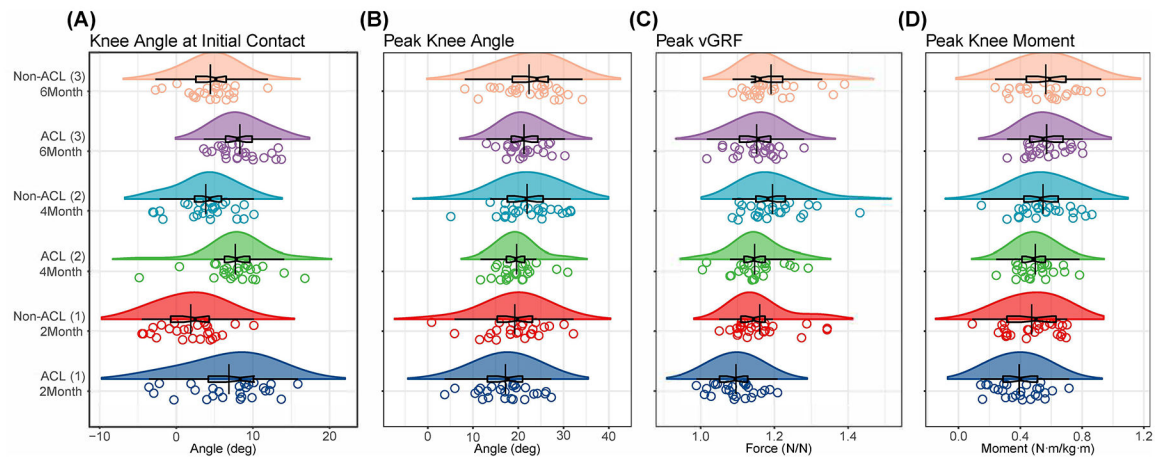


Figure 2:

Raincloud plots showing distributions of (A) knee flexion angle at initial contact, (B) peak knee flexion angle, (C) peak vertical ground reaction force (vGRF), and (D) peak knee extension moment during early stance for the ACL-reconstructed limb (ACL) and non-ACL-reconstructed limb (Non-ACL) at the three timepoints (about 2 months, 4 months, and 6 months post-surgery) tested in this study. The circles on the plot indicate individual data points for the reconstructed and the non-reconstructed limbs at timepoint 1 (1), timepoint 2 (2), and timepoint 3 (3). Rainclouds and notched box plots show the distribution of the data, and the horizontal line on the box plot indicates the mean of the distribution. Note that a lower knee angle at initial contact (A) indicates that the participant walked with a more extended knee at heel strike, and a lower peak knee angle (B) indicates that the participant walked with reduced knee flexion during the early stance phase.

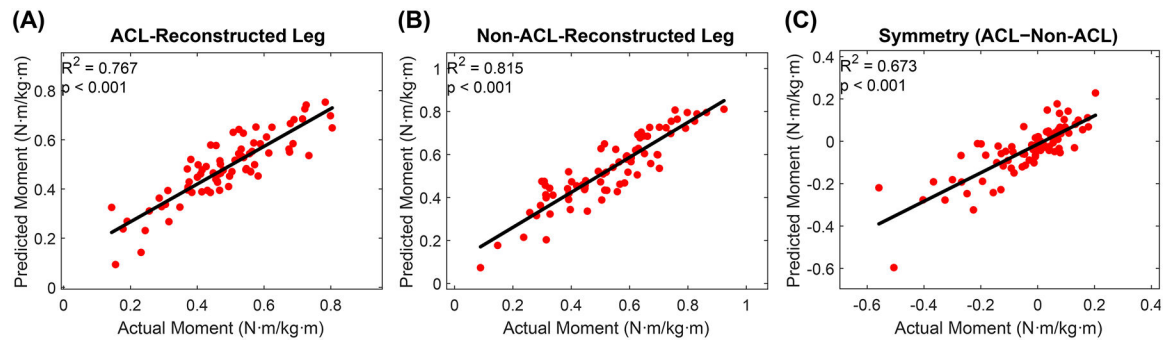


Figure 3:

(A) A scatterplot of actual vs. predicted knee moments of the ACL-reconstructed limb (ACL) using the regression model that used peak knee flexion angle and peak vGRF during early stance in the ACL-reconstructed limb. (B) A scatterplot of actual vs. predicted knee moments of the non-ACL-reconstructed limb (Non-ACL) using the regression models that used peak knee flexion angle and peak vGRF during early stance in the non-ACL-reconstructed limb. (C) A scatterplot of actual vs. predicted knee moment symmetry (*i.e.*, ACL limb values–non-ACL limb values) using the regression model that used peak knee flexion angle and peak vGRF symmetry values during early stance.

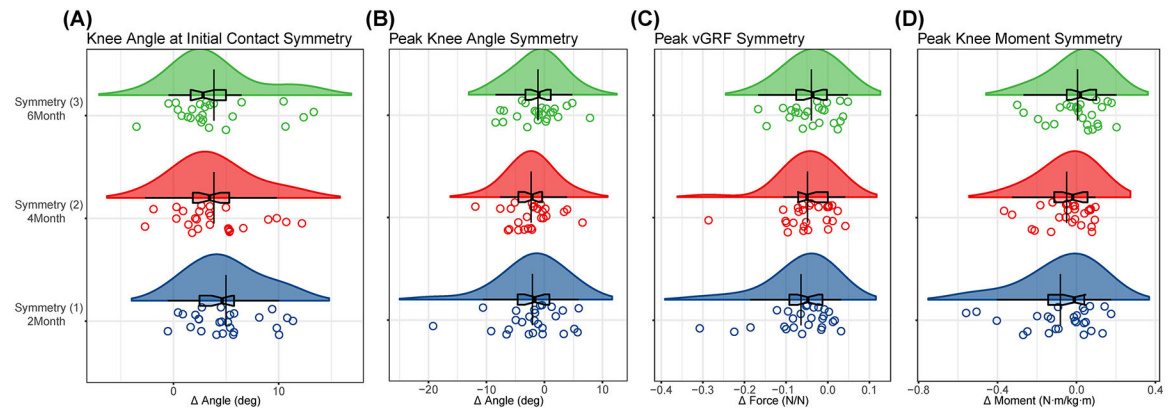


Figure 4:

Raincloud plots showing distributions of symmetry values (i.e., ACL-reconstructed limb values–non-ACL-reconstructed limb values) for (A) knee flexion angle at initial contact, (B) peak knee flexion angle, (C) peak vertical ground reaction force (vGRF), and (D) peak knee extension moment during early stance for the ACL-reconstructed limb (ACL) and non-ACL-reconstructed limb (Non-ACL) at the three timepoints (about 2 months, 4 months, and 6 months post-surgery) tested in this study. The circles on the plot indicate individual data points for the reconstructed and the non-reconstructed limbs at timepoint 1 (1), timepoint 2 (2), and timepoint 3 (3). Rainclouds and notched box plots show the distribution of the data, and the horizontal line on the box plot indicates the mean of the distribution. Note that a negative Δ value in limb symmetry indicates that the ACL-reconstructed limb had a lower value than the non-ACL-reconstructed limb, and vice versa for a positive Δ value.

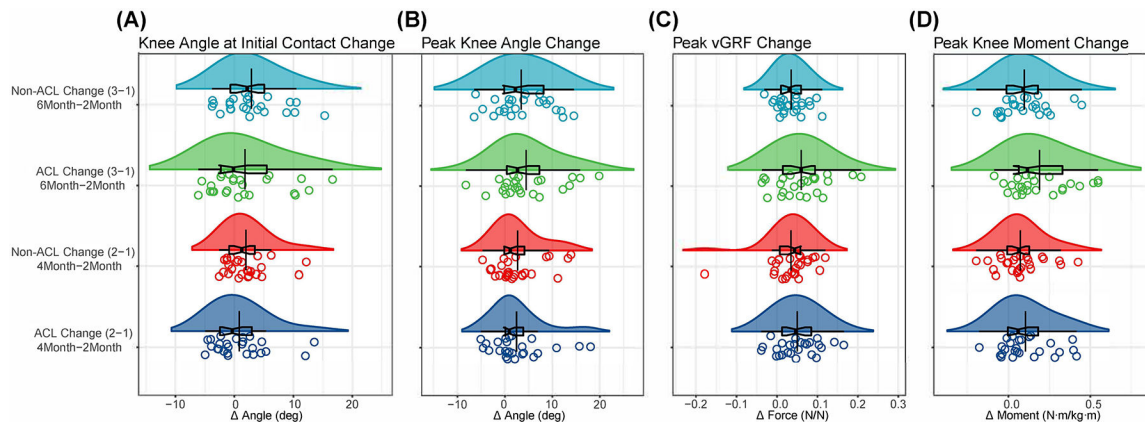


Figure 5:

Raincloud plots showing distributions of change scores (i.e., changes across timepoints) for (A) knee flexion angle at initial contact, (B) peak knee flexion angle, (C) peak vertical ground reaction force (vGRF), and (D) peak knee extension moment during early stance for the ACL-reconstructed limb (ACL) and non-ACL-reconstructed limb (Non-ACL) at the three timepoints (about 2 months, 4 months, and 6 months post-surgery) tested in this study. The circles on the plot indicate individual data points for the reconstructed and the non-reconstructed limbs at timepoint 2 (2-1) and timepoint 3 (3-1), and timepoint 3 (3). Rainclouds and notched box plots show the distribution of the data, and the horizontal line on the box plot indicates the mean of the distribution. Note that a negative Δ value in the change score indicates that the baseline value (timepoint 1) had a higher value than timepoint 2 or 3, and vice versa for a positive Δ value.

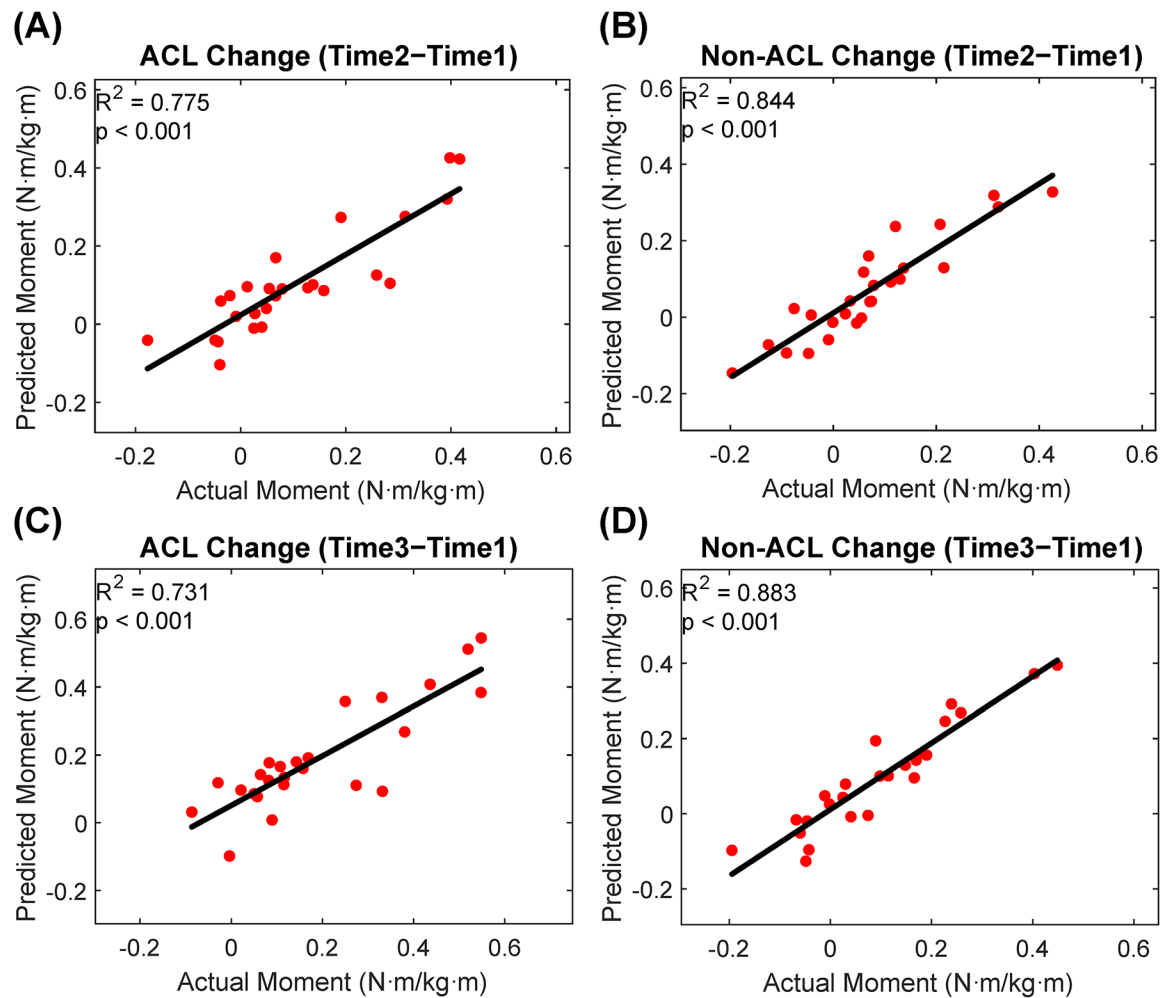


Figure 6:

(A) Scatterplots of actual vs. predicted changes in knee moment of the ACL-reconstructed limb (ACL) and non-ACL-reconstructed limb (Non-ACL) using regression models that used changes in knee flexion angle at initial contact, peak knee flexion angle, and peak vGRF during early stance of the respective limbs across timepoints. Note that the ACL-reconstructed limb had peak knee flexion angle and peak vGRF as predictors in the model whereas the non-ACL-reconstructed limb had knee flexion angle at initial contact, peak knee flexion angle, and peak vGRF as predictors in the model.

Table 1 –

Interpretation of Bayes Factor (BF_{10}) in favor of alternative hypothesis/model (H_1) over null hypothesis/model (H_0).

Bayes Factor (BF_{10})	Interpretation
>100	Decisive evidence favoring the alternative hypothesis/model
$30-100$	Very strong evidence favoring the alternative hypothesis/model
$10-30$	Strong evidence favoring the alternative hypothesis/model
$3-10$	Moderate evidence favoring the alternative hypothesis/model
$1-3$	Anecdotal evidence favoring the alternative hypothesis/model
1	No evidence favoring the null or the alternative hypothesis/model
$1/3-1$	Anecdotal evidence favoring the null hypothesis/model
$1/10-1/3$	Moderate evidence favoring the null hypothesis/model
$1/30-1/10$	Strong evidence favoring the null hypothesis/model
$1/100-1/30$	Very strong evidence favoring the null hypothesis/model
$<1/100$	Decisive evidence favoring the null hypothesis/model